Activity - 7

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correlation detective online store case

1) correlation matrix

Number of observation: $n=17$

no-of variables $=7$

Distinct variables pairs $= \frac{7(7-1)}{2} = 21$

round(cor(case),2)

2) Strongest correlations

Strongest positive :- festive-Index and ad clicks,

$r = 0.97$.

Strongest negative :- festive-Index & parcels $r = -0.92$

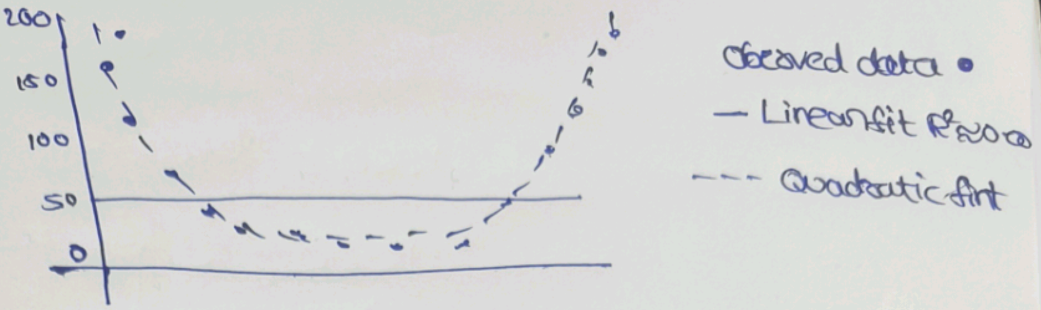
Both are strong correlations.

3) Deceptive zero :-

Straight-line fit $R^2 \approx 0.00$

Quadratic fit $R^2 \approx 0.99$

correlation misses the curved relationship b/w delay_dev & complaints.



4) Outlier Alibi :-

Wrong record = row-6

Staff online = 29; returns = 51

correlation with all rows : $r = -0.68$

without row 6: $r = 0.02$

Row 6 is outlier that changes the correlation.

5) Prime suspect :-

=> correlation b/w ad clicks and parcels $r \approx -0.90$

=> festive Index and ad clicks $r \approx 0.97$

=> festive Index and parcels $r \approx -0.92$

=> partial correlation controlling for festival Index

$r \approx -0.07$